

LAB No. 20

Backup recovery 5-day rotation policy using Cronjob

In Linux systems, regular backups are essential to protect data from accidental deletion, system failure, or cyber incidents. However, uncontrolled backups can consume excessive storage space. Backup rotation is a technique used to manage storage efficiently by retaining backups for a fixed number of days and automatically removing older backups.

This lab focuses on developing a **Linux shell script that performs automated backups with a 5-day rotation policy**. The script creates daily backups of selected directories and retains only the most recent five backup copies, automatically deleting older backups. The backup process is automated using **cron jobs**, ensuring reliable and space-efficient data protection.

LAB Objectives

- Understand backup rotation concepts
- Create automated backup scripts using shell scripting
- Implement a 5-day backup retention policy
- Automatically delete old backup files
- Schedule backups using cron jobs
- Maintain logs for backup activities

Tools and Software Required

Operating System

- Linux (Ubuntu / CentOS / Red Hat / Debian)

Backup Rotation Policy

Backup rotation limits the number of stored backups by automatically deleting older ones. A **5-day rotation** keeps only the last five daily backups, balancing data safety and storage usage.

Benefits:

- Prevents disk space exhaustion
- Ensures recent backups are always available

- Reduces manual cleanup

Shell Scripting for Backup Automation

Shell scripts automate:

- Backup creation
- File naming with date and time
- Rotation and cleanup of old backups
- Error handling and logging

Automation ensures consistency and reliability.

Compression and Archiving

Compression reduces backup size and speeds up storage and transfer. The tar command combines multiple files, while gzip compresses them efficiently.

Cron Job Scheduling

Cron schedules the backup script to run automatically at fixed intervals such as daily or nightly, ensuring backups are taken regularly without user intervention.

Expected Outcomes

- Automated backup script with 5-day retention
- Scheduled backup execution via cron job
- Automatic deletion of backups older than 5 days
- Verified backup logs

Lab Task

Perform the following tasks:

1. Create a shell script to back up a specified directory using tar.
2. Name backup files using the current date.
3. Store backup files in a dedicated backup directory.
4. Implement a 5-day rotation policy to automatically delete older backups.
5. Add logging to record backup and rotation activities.
6. Schedule the script using a cron job to run daily.
7. Verify that only the last five backups are retained.

OUTPUT:

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ mkdir -p ~/backups  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ nano backup.sh  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat backup.sh  
#!/bin/bash  
  
# SOURCE DIRECTORY TO BACKUP  
SOURCE_DIR="$HOME/Documents"  
  
# BACKUP DIRECTORY  
BACKUP_DIR="$HOME/backups"  
  
# LOG FILE  
LOG_FILE="$BACKUP_DIR/backup.log"  
  
# CURRENT DATE  
DATE=$(date +%Y-%m-%d_%H-%M-%S)  
  
# BACKUP FILE NAME  
BACKUP_FILE="$BACKUP_DIR/backup_$DATE.tar.gz"  
  
# CREATE BACKUP DIRECTORY IF NOT EXISTS  
mkdir -p "$BACKUP_DIR"  
  
# CREATE BACKUP  
tar -czf "$BACKUP_FILE" "$SOURCE_DIR"  
  
# CHECK IF BACKUP SUCCESSFUL  
if [ $? -eq 0 ]; then  
    echo "$(date) : Backup created successfully -> $BACKUP_FILE" >> "$LOG_FILE"  
else  
    echo "$(date) : Backup failed!" >> "$LOG_FILE"  
fi
```

```
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~  
  
# DELETE BACKUPS OLDER THAN 5 DAYS  
find "$BACKUP_DIR" -name "*.tar.gz" -type f -mtime +5 -exec rm -f {} \;  
  
# LOG CLEANUP  
echo "$(date) : Old backups deleted (older than 5 days)" >> "$LOG_FILE"  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ nano backup.sh  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ chmod +x backup.sh  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ./backup.sh  
tar: Removing leading '/' from member names  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/backups  
backup_2026-05-19_07-17-07.tar.gz  backup.log  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ cat ~/backups/backup.log  
Tue May 19 07:17:07 AM PKT 2026 : Backup created successfully -> /home/abdullah-sohail/backups/backup_2026-05-19_07-17-07.tar.gz  
Tue May 19 07:17:07 AM PKT 2026 : Old backups deleted (older than 5 days)  
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ crontab -e  
crontab: installing new crontab
```

```

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform: ~
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
*/1 * * * * /bin/bash /home/abdullah-sohail/health_monitor.sh
0 * * * * /home/abdullah-sohail/disk_monitor.sh
0 0 * * * /home/abdullah-sohail/backup.sh
0 0 * * * /home/abdullah-sohail/backup.sh
0 1 * * * /home/abdullah-sohail/backup.sh

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/backups
backup_2026-05-19_07-17-07.tar.gz  backup.log
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$

abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ ls ~/backups
backup_2026-05-19_07-17-07.tar.gz  backup.log
abdullah-sohail@abdullah-sohail-VMware-Virtual-Platform:~$ tar -tzf ~/backups/backup_2026-05-19_07-17-07.tar.gz
home/abdullah-sohail/Documents/

```